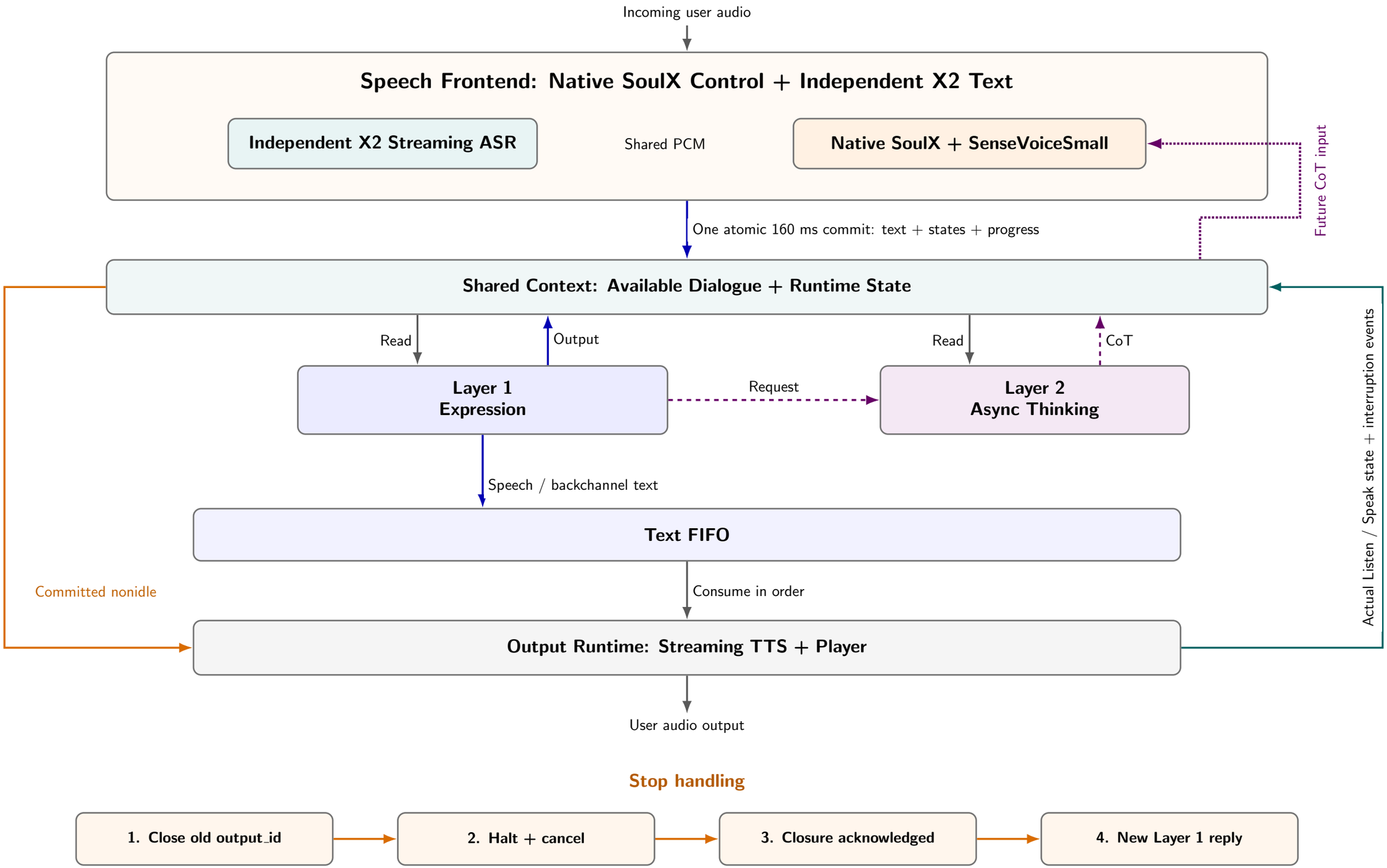
ParaThink: Speech Frontend, Parallel Thinking and Expression
Incoming user audio
Speech Frontend: Native SoulX Control + Independent X2 Text
Independent X2 Streaming ASR
Shared PCM
Native SoulX + SenseVoiceSmall
Future CoT input
One atomic 160 ms commit: text + states + progress
Shared Context: Available Dialogue + Runtime State
Read
Output
Read
CoT
Layer 1
Expression
Request
Layer 2
Async Thinking
Speech / backchannel text
Text FIFO
Committed nonidle
Consume in order
Output Runtime: Streaming TTS + Player
Actual Listen / Speak state + interruption events
User audio output
Stop handling
1. Close old output_id
2. Halt + cancel
3. Closure acknowledged
4. New Layer 1 reply